Precision Placement in Scaled Dot-Product Attention

Ján Kovačovský

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Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We study scaled dot-product attention as a precision-placement problem. Rather than asking whether a single global low-precision dtype is acceptable, we ask which stages of the forward pass can safely use reduced precision: storage of query, key, and value tensors, the logit product, logit storage, rowwise softmax, and the final value product. We prove a local perturbation bound separating logit errors from value errors, and we extend the same viewpoint to quantization by modeling quantized tensors as structured perturbations. Exact-attention experiments show that low-precision storage with high-precision accumulation is comparatively stable, whereas low-precision softmax is the clearest floating-point failure mode. A temperature-scaled attention experiment reveals a finite-size entropy-collapse crossover in which static int8 logit quantization fails through clipping and argmax flips. Finally, an Ising magnetization task gives a preliminary training-time test of whether reduced precision or randomized sketching changes learnability. The sketched-attention baseline is kept as a controlled stress test for the interaction between randomized approximation and finite precision.

Keywords: mixed-precision arithmetic, attention, transformers, numerical stability, floating-point arithmetic

1 Introduction

Scaled dot-product attention [1] is the core computational primitive of transformer architectures. As models grow, the memory and compute cost of the attention forward pass motivates the use of reduced-precision arithmetic. The natural question is a *precision-placement* question: which stages of the forward pass can safely run in low precision, and which still require higher-precision accumulation or reduction?

The relevant stages are: storage of the query, key, and value tensors; the logit product QK^T; logit storage; the rowwise softmax; and the final value product. A global dtype comparison—the usual approach—conflates these stages and cannot identify which *locations* are numerically sensitive. This report instead decomposes the attention pipeline and asks, for each stage separately, whether low precision there produces acceptable error.

The analytical core is Proposition 3, a local perturbation bound for exact attention that separates the effect of logit perturbations (weighted by $\|V\|_2$) from value perturbations (weighted by $\sqrt{n}$ from the row-stochastic spectral norm). This decomposition directly predicts which precision locations are sensitive: logit errors propagate through softmax with unit Lipschitz constant, while value errors propagate with a $\sqrt{n}$ factor that is moderate for the small sequence lengths tested here, but potentially important for long contexts. Among the floating-point placement policies, low-precision softmax is the clearest failure mode. Under explicit quantization, however, static logit quantization can fail more severely through clipping once the attention temperature moves outside the calibration range.

Sketching-based attention approximations [2] [3] [4] [5] [6] are retained as a methodological baseline: they provide a controlled stress test in which the error decomposition can be verified under a known approximation floor. But they are not the main contribution.

Although the experiments below focus mainly on floating-point precision policies, the same placement question also covers quantization. In quantized inference, the choice is not only which dtype to use, but also which tensors to quantize, how to choose scales and zero-points, whether calibration is dynamic or static, and which reductions must remain in higher precision. We therefore treat quantization as an extension of the same local-perturbation framework.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n\times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \to \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q, K) := d^{-1/2} QK^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

3 Analytical Core

3.1 Mixed Precision versus Quantization

In this report we distinguish mixed precision from quantization. Mixed precision changes the floating-point format used at different stages of a computation, for example storing Q, K, V in `float16` or `bfloat16` while accumulating QK^T and computing the softmax in `float32`. Quantization is more explicit: a real-valued tensor X is mapped to a discrete low-bit representation and then, for computation, usually dequantized back to a floating-point format.

A common affine quantization model is

$$x \approx S(x_q - Z),$$

where x_q is the quantized integer value, $S > 0$ is a scale, and Z is a zero-point chosen so that real zero can be represented exactly. Equivalently,

$$x_q = \operatorname{clip}(\operatorname{round}(x/S + Z)).$$

Symmetric quantization corresponds to the special case $Z = 0$ after choosing a symmetric representable range. The scale and zero-point may be chosen per tensor or per channel, and activation ranges may be determined dynamically, by static calibration, or during quantization-aware training.

For attention, quantization is another precision-placement mechanism. One may quantize Q, K, V, the logits $L = QK^T/\sqrt{d}$, the attention probabilities $\Sigma(L)$, or the final output. The perturbation framework below treats quantization by writing

$$\widehat{X} = X + \Delta_{\text{quant}} X$$

for each quantized tensor, where $\Delta_{\text{quant}} X = D(Q(X)) - X$. Thus quantization errors enter the same local error bound as floating-point storage and rounding errors, but their structure is different: they depend on calibration range, clipping, scale granularity, and dequantization policy.

Mechanism	Typical form	Main error source
Mixed precision	fp16/bf16/fp8 storage, fp32 accumulation	floating-point rounding and overflo
Affine quantization	$x \approx S(x_q - Z)$	rounding, clipping, calibration erro
Symmetric quantization	$x \approx Sx_q$	rounding and clipping under symm
Dynamic activation quantization	runtime range estimation	range-estimation overhead and var
Static activation quantization	calibrated range from representative data	calibration mismatch
Quantization-aware training	fake quantization during training	training adapts to quantization no

Table 1 / Mixed precision and quantization as precision-placement mechanisms.

The analysis begins with a stability property of the rowwise softmax map, which is used in both the exact-attention and sketching bounds.

Proposition 2 (Rowwise softmax is Hilbert–Schmidt Lipschitz)**.** Let $L, M \in \mathbb{R}^{n\times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \le \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n\times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \le \|L - M\|_{\text{HS}}\, \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \operatorname{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x)\, z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \le \|z\|_2^2$, so $\|D\sigma(x)\|_2 \le 1$ for all x. The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \le \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\mathrm{HS}} \leq \|X\|_{\mathrm{HS}}\|V\|_2$. □

3.2 Exact-Attention Perturbation Bound

The next proposition is the main analytical result. It decomposes the attention error into a logit-driven term and a value-driven term, directly predicting the relative sensitivity of different precision placements.

Proposition 3 (Exact-attention local perturbation bound)**.** Let

$$A = \Sigma(L)V, \qquad \widehat{A} = \Sigma(\widehat{L})\widehat{V},$$

where $L, \widehat{L} \in \mathbb{R}^{n\times n}$ and $V, \widehat{V} \in \mathbb{R}^{n\times d}$. Writing

$$\Delta L := \widehat{L} - L, \qquad \Delta V := \widehat{V} - V,$$

one has

$$\|A - \widehat{A}\|_{\mathrm{HS}} \leq \|\Delta L\|_{\mathrm{HS}}\, \|V\|_2 + \|\Sigma(\widehat{L})\|_2\, \|\Delta V\|_{\mathrm{HS}}.$$

In particular, since $\Sigma(\widehat{L})$ is row-stochastic and therefore $\|\Sigma(\widehat{L})\|_2 \leq \sqrt{n}$,

$$\|A - \widehat{A}\|_{\mathrm{HS}} \leq \|\Delta L\|_{\mathrm{HS}}\, \|V\|_2 + \sqrt{n}\, \|\Delta V\|_{\mathrm{HS}}.$$

Proof. Add and subtract $\Sigma(\widehat{L})V$ and apply the triangle inequality:

$$\|A - \widehat{A}\|_{\mathrm{HS}} \leq \|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} + \|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}}.$$

The first term is bounded by Proposition 2:

$$\|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} \leq \|L - \widehat{L}\|_{\mathrm{HS}}\, \|V\|_2.$$

For the second,

$$\|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}} \leq \|\Sigma(\widehat{L})\|_2\, \|\Delta V\|_{\mathrm{HS}}.$$

Combining these gives the stated bound. □

Remark 4 (Interpretation for precision placement)**.** Proposition 3 directly predicts the relative danger of different precision choices:

(1) **Logit perturbations** ($\Delta L \neq 0$, $\Delta V = 0$): The error scales as $\|\Delta L\|_{\mathrm{HS}}\|V\|_2$. Exact softmax is Lipschitz and therefore does not amplify logit perturbations in the HS/L2 sense. However, the diagnostic ratio ρ_{softmax}, which compares a rowwise ℓ_∞ logit error to an L1 probability error, can become large when the logit perturbation denominator is extremely small or when the softmax computation itself is rounded in low precision—a regime the experiments expose.

(2) **Value perturbations** ($\Delta L = 0$, $\Delta V \neq 0$): The error scales as $\sqrt{n}\, \|\Delta V\|_{\mathrm{HS}}$. This $\sqrt{n}$ factor is moderate for the small sequence lengths tested here, but potentially important for long contexts.

(3) **Low-precision softmax**: When Σ itself is computed in low precision, the Lipschitz bound from Proposition 2 no longer applies, and the effective perturbation of the probability matrix can be much larger than $\|\Delta L\|_{\mathrm{HS}}$. Low-precision softmax thus introduces additional probability-matrix error beyond the perturbation predicted by exact-softmax Lipschitz stability. The experiments confirm this: ρ_{softmax} spikes to $\sim 10^3$ on transformer activations when softmax is run in bf16/fp16, while remaining below 1 on Gaussian data and only order-unity for safe transformer policies.

3.3 Supporting Background: Sketching as a Controlled Baseline

The following propositions analyze a sketched surrogate $\widetilde{\mathcal{A}}_S(Q,K,V) = \Sigma(d^{-1/2}QSS^*K^*)V$ obtained by replacing the full logit product with a low-dimensional projection through $S \in \mathbb{R}^{d\times s}$ with $s < d$. These results are retained as a methodological baseline: the sketching error provides a known approximation floor against which the finite-precision error can be isolated and studied.

Proposition 5 (Deterministic sketching bound)**.** Let $U \subseteq \mathcal{H}_d$ be the joint row space of Q and K, equivalently the subspace spanned by the columns of $[Q^T \;\; K^T]$, and let $r := \dim U$. Assume that S is an ϵ-subspace embedding for U, meaning

$$\left|\langle S^*u, S^*v\rangle - \langle u, v\rangle\right| \le \epsilon\,\|u\|_2\,\|v\|_2 \qquad \text{for all } u, v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d\times r}$ is any orthonormal basis matrix for U, then $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$. Then

$$\|\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\,\|Q\|_{\mathrm{HS}}\,\|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q,K,V) - \widetilde{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\,\|Q\|_{\mathrm{HS}}\,\|K\|_2\,\|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ have orthonormal columns spanning the joint row space of Q and K. Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = QU_0 \in \mathbb{R}^{n\times r}$ and $K_U = KU_0 \in \mathbb{R}^{n\times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T SS^*U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\mathrm{HS}} \le \|Q_U\|_{\mathrm{HS}}\,\|I_r - U_0^T SS^*U_0\|_2\,\|K_U^T\|_2 \le \epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2,$$

where we used $\|Q_U\|_{\mathrm{HS}} = \|Q\|_{\mathrm{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 2. □

Proposition 6 (Probabilistic subspace embedding guarantee)**.** Let $U \subseteq \mathcal{H}_d$ be an r-dimensional subspace. Let $S \in \mathbb{R}^{d\times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For

$\epsilon \in (0,1)$ and $\delta \in (0,1)$, if

$$s \;\geq\; C\,\frac{r + \ln(1/\delta)}{\epsilon^2}\,,$$

for a universal constant C, then with probability at least $1-\delta$, S is an ϵ-subspace embedding for U.

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ be a matrix whose columns form an orthonormal basis for U. Then S is an ϵ-subspace embedding for U if and only if $\|U_0^T SS^* U_0 - I_r\|_2 \leq \epsilon$. The matrix $U_0^T SS^* U_0 = (S^*U_0)^*(S^*U_0)$ has the same distribution as G^*G/s where $G \in \mathbb{R}^{s\times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives $\Pr(\|G^*G/s - I_r\|_2 > \epsilon) \leq 2\exp(-c\,s\epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2. The stated condition with an appropriate C ensures the failure probability is at most δ. □

Proposition 7 (Finite-precision error for sketched logits)**.** Let $\widehat{\mathcal{L}}_S(Q,K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u, and let $\gamma_s := su/(1-su)$ for $su < 1$. Then

$$\|\widetilde{\mathcal{L}}_S(Q,K) - \widehat{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \leq \frac{\gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}}{\sqrt{d}}\,.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n\times s}$ and $B = KS \in \mathbb{R}^{n\times s}$. The standard forward-error bound for matrix multiplication [7] gives

$$|\widehat{AB}^* - AB^*| \leq \gamma_s\,|A|\,|B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB}^* - AB^*\|_{\mathrm{HS}} \leq \gamma_s\,\||A|\,|B^*|\|_{\mathrm{HS}} \leq \gamma_s\,\|A\|_{\mathrm{HS}}\,\|B\|_{\mathrm{HS}}.$$

Multiplying by $d^{-1/2}$ yields the bound. □

Remark 8 (Scope of the finite-precision bound)**.** The bound in Proposition 7 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error that also includes storage quantization and low-precision projection effects. ⌟

Combining the subspace embedding guarantee with the finite-precision error yields the sketching main theorem.

Theorem 9 (Mixed-precision sketched attention)**.** Under the hypotheses of Propositions 6 and 7, with probability at least $1-\delta$,

$$\|\mathcal{A}(Q,K,V) - \widehat{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \leq \frac{1}{\sqrt{d}}\Big(\epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2 + \gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}\Big)\|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error.

Proof. By Proposition 2 and the triangle inequality,

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\mathrm{HS}} \le (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}})\, \|V\|_2.$$

Apply Proposition 5 to the first term and Proposition 7 to the second. □

Remark 10 (Design rule from the sketching theorem)**.** Theorem 9 yields a concrete joint design rule. For float16 ($u_{16} \approx 5 \times 10^{-4}$) and typical sketch dimensions $s \le 64$, we have $s \cdot u_{16} \le 0.032$, which is smaller than typical sketch tolerances $\epsilon \ge 0.1$. For float32 ($u_{32} \approx 6 \times 10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\mathrm{fp8}} \approx 0.0625$), even $s = 16$ gives $s \cdot u_{\mathrm{fp8}} \approx 1.0$, which is comparable to ϵ—so the finite-precision term is no longer clearly lower order. ⌟

4 Numerical Experiments

The experiments are organized in five layers. Section 4.1 studies exact attention under fixed precision-placement policies. Section 4.2 introduces a temperature parameter to expose an entropy-collapse crossover and quantization clipping. Section 4.3 studies how local precision errors accumulate in a toy residual stack. Section 4.4 uses an Ising magnetization task as a training-time sanity check and masked-observation stress test. Section 4.5 returns to sketched attention as an intentionally under-sketched baseline for the error decomposition.

4.1 Exact Attention Under Precision Policies

For each precision policy, we evaluate the exact attention pipeline

$$Q, K, V \to L = QK^*/\sqrt{d} \to P = \Sigma(L) \to A = PV$$

against a `float64` reference. The primary diagnostics are the relative logit error

$$E_L = \frac{\|L - \widehat{L}\|_{\mathrm{HS}}}{\|L\|_{\mathrm{HS}}},$$

the mean rowwise softmax error

$$E_P = \frac{1}{n} \sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1,$$

the relative output error

$$E_A = \frac{\|A - \widehat{A}\|_{\mathrm{HS}}}{\|A\|_{\mathrm{HS}}},$$

and the softmax amplification ratio

$$\rho_{\mathrm{softmax}} = \frac{\frac{1}{n}\sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1}{\frac{1}{n}\sum_{i=1}^{n} \|\ell_i - \widehat{\ell}_i\|_\infty}.$$

We test the following policy family: `fp32_reference`, `bf16_safe`, `fp16_safe`, low-precision logit storage, low-precision softmax, and low-precision value accumulation. The "safe" policies keep storage low precision but preserve fp32 accumulation and fp32 softmax. This separation is the core experimental point: the question is not whether one global dtype works, but which *locations* are sensitive. The `fp8_store` policy in the sketching baseline (Table 19) should be interpreted as a floating-point quantization-like storage policy rather than affine integer quantization: it reduces representational precision, but it does not introduce an explicit scale and zero-point. In contrast, an int8 attention policy would require choosing calibration ranges for Q, K, V, possibly per tensor or per channel, and would introduce clipping error in addition to floating-point rounding.

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	8.2e-8	8.5e-8	1.4e-7	0.41
`bf16_safe`	2.4e-3	1.9e-3	2.8e-3	0.35
`fp16_safe`	2.9e-4	2.1e-4	3.2e-4	0.32
`bf16_low_logit`	2.9e-3	2.3e-3	3.4e-3	0.34
`fp16_low_logit`	3.5e-4	2.8e-4	4.4e-4	0.33
`bf16_low_softmax`	2.9e-3	2.7e-3	3.7e-3	0.41
`fp16_low_softmax`	3.5e-4	3.3e-4	4.7e-4	0.39
`bf16_low_value`	2.4e-3	1.9e-3	3.5e-3	0.35
`fp16_low_value`	2.9e-4	2.1e-4	4.4e-4	0.32

Table 11 / Exact-attention errors on synthetic Gaussian data ($n \in \{16, 32\}$, $d = 16$, averaged over 2 seeds). "Safe" policies use low-precision storage with fp32 accumulation and softmax.

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	1.8e-8	3.0e-8	5.0e-8	4.6e4
`bf16_safe`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_safe`	3.3e-4	3.0e-8	3.5e-4	3.4
`bf16_low_logit`	3.9e-4	3.0e-8	1.8e-3	2.8
`fp16_low_logit`	3.9e-4	3.0e-8	3.5e-4	2.8
`bf16_low_softmax`	3.9e-4	1.2e-5	1.8e-3	1.1e3
`fp16_low_softmax`	3.9e-4	1.2e-5	3.4e-4	1.1e3
`bf16_low_value`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_low_value`	3.3e-4	3.0e-8	3.4e-4	3.4

Table 12 / Smoke-test exact-attention errors on a single attention head extracted from a tiny pretrained model (`sshleifer/tiny-distilbert-base-cased`, layer 0, head 0; $n = 16$, $d = 1$). Because the head dimension is degenerate, this table serves as a consistency check rather than full real-activation evidence. The softmax amplification ratio ρ_{softmax} spikes under low-precision softmax, consistent with Proposition 3 and Remark 4.

Tables 11 and 12 report representative policy-level numbers from Gaussian and transformer-activation sweeps respectively. On Gaussian data (Table 11), all policies remain low-error because the softmax is not in a numerically sensitive regime: $\rho_{\text{softmax}} < 1$ uniformly, so probability errors are generally damped relative to rowwise logit perturbations. The output error E_A remains of the same order as the local logit/value errors. The differences between "safe" and "low" variants are modest in this regime.

The transformer activations (Table 12) reveal a qualitatively different regime. The baseline ρ_{softmax} for `fp32_reference` is $\sim 10^4$ because both numerator and denominator are extremely small, but the rowwise logit-error denominator is several orders of magnitude smaller than the probability-error numerator. This is a diagnostic artifact of dividing two near-zero quantities. The important comparison is between "safe" policies ($\rho \sim$ 1–3) and "low_softmax" policies ($\rho \sim 10^3$). The low-softmax rows should be interpreted relative to the corresponding low-logit rows: both store logits in low precision, but only the low-softmax policy also performs the softmax reduction in low precision. This difference is enough to raise E_P by two orders of magnitude relative to the safe baseline, confirming that low-precision softmax is the clearest failure mode at the probability-matrix level.

The most important empirical message across both tables is that low-precision softmax is the clearest failure mode at the probability-matrix level, while in the tested small regimes, low-precision storage of Q, K, V with fp32 accumulation remains much safer than low-precision softmax at the probability-matrix level. This is precisely the kind of location-specific statement that a global dtype comparison would miss. The full exact-attention and residual-depth sweeps are committed in `results/precision_policy_sweep.dev.csv`, `results/precision_policy_transformer.dev.csv`, `results/residual_stack_sweep.dev.csv`, and `results/residual_stack_transformer.dev.csv`.

4.2 Attention Entropy Transition and Quantization

A useful way to expose precision sensitivity inside the attention block itself is to introduce an inverse-temperature parameter β and study

$$P_\beta = \operatorname{softmax}(\beta L), \qquad L = QK^*/\sqrt{d}.$$

For small β, the attention rows are diffuse and high-entropy. For large β, the rows become concentrated and approach a winner-take-all map. For finite sequence length this is not a thermodynamic phase transition, but it is a sharp finite-size crossover in the geometry of the attention matrix.

We track the normalized entropy

$$\frac{H_\beta}{\log n} = -\frac{1}{n \log n} \sum_{i,j} p_{ij}^{(\beta)} \log p_{ij}^{(\beta)},$$

the mean maximum attention mass

$$M_\beta = \frac{1}{n} \sum_i \max_j p_{ij}^{(\beta)},$$

the probability error E_P, the output error E_A, and the argmax flip rate

$$R_{\text{flip}} = \frac{1}{n} \#\{i : \arg\max_j p_{ij}^{(\beta)} \neq \arg\max_j \widehat{p}_{ij}^{(\beta)}\}.$$

This experiment also clarifies the difference between mixed precision and quantization. Mixed precision changes the floating-point format used at different stages of the computation. Quantization introduces an explicit map from real values to a discrete low-bit representation. In the symmetric int8 model used here,

$$q = \operatorname{clip}(\operatorname{round}(x/S), -127, 127), \qquad \widehat{x} = Sq.$$

Dynamic quantization chooses S from the current tensor range, while the static-logit policy fixes the scale using the $\beta = 1$ logits and reuses it for all β. The latter deliberately exposes clipping once the inverse temperature becomes large. In attention, quantization can be placed on Q, K, V, on the logits, on the probabilities, or on the output; here we test QKV quantization and logit quantization.

Policy / β	$H/\log n$	M_β	E_P	E_A	R_{flip}	Clip
`fp16_safe`, $\beta = 0.5$	0.975	0.025	1.12e-4	2.64e-4	0.000	—
`fp16_safe`, $\beta = 4$	0.335	0.534	6.72e-4	6.80e-4	0.000	—
`fp16_safe`, $\beta = 16$	0.062	0.883	7.83e-4	1.15e-3	0.000	—
`fp16_low_softmax`, $\beta = 0.5$	0.975	0.025	2.23e-4	3.65e-4	0.004	—
`fp16_low_softmax`, $\beta = 4$	0.335	0.534	1.24e-3	1.27e-3	0.004	—
`fp16_low_softmax`, $\beta = 16$	0.062	0.883	1.55e-3	2.11e-3	0.004	—
`int8_logits_static_beta1`, $\beta = 0.5$	0.975	0.025	7.56e-3	8.66e-3	0.047	0.000
`int8_logits_static_beta1`, $\beta = 4$	0.335	0.534	1.44e+0	9.52e-1	0.922	0.322
`int8_logits_static_beta1`, $\beta = 16$	0.062	0.883	1.92e+0	9.92e-1	0.980	0.804

Table 13 / Representative attention entropy-transition results on Gaussian data ($n = 128$, $d = 32$, averaged over 2 seeds). The full sweep is committed in `results/attention_temperature_sweep.dev.csv`.

Table 13 shows representative rows from the entropy-transition sweep. As β increases, $H/\log n$ decreases from near-unity to near-zero while M_β increases from near-$1/n$ to near-unity: this is the finite-size crossover from diffuse to concentrated attention. Safe floating-point policies remain stable through this transition: E_P and E_A increase by less than an order of magnitude and R_{flip} stays at zero. Low-precision softmax shows modestly elevated errors but no catastrophic breakdown.

The static int8 logit policy tells a different story. At $\beta = 0.5$ the logits lie within the calibration range inherited from $\beta = 1$, clipping is zero, and errors are small. By $\beta = 4$ the rescaled logits βL exceed the calibration range for roughly 32% of entries, E_P exceeds unity, and R_{flip} reaches 92%. At $\beta = 16$ clipping affects 80% of entries and the argmax flip rate is 98%—the quantized attention pattern has almost no overlap with the reference. This is precisely the fragility predicted by the quantization model: once the scale is miscalibrated, clipping introduces large structured errors that the perturbation bound in Proposition 3 does not capture, because $\Delta_{\text{quant}} L$ is no longer small.

The transition experiment thus gives a concrete phase-diagram view of precision placement: low precision is easiest in the diffuse regime and most fragile near concentrated, winner-take-all attention. This is the clearest phase-diagram behavior in the current report: as β drives attention through the entropy-collapse crossover, the fixed int8 calibration range becomes invalid, clipping appears abruptly, and argmax attention decisions fail almost completely.

4.3 Residual Depth Propagation

To test how local precision errors accumulate across depth, we evaluate a toy attention-only residual dynamical system of the form

$$x_{\ell+1} = x_\ell + hF_\ell(x_\ell)$$

with self-attention blocks F_ℓ. This is a toy stability test—not a full transformer block, since it has no learned projections, MLP, or layer norm—but it isolates how the residual mechanism propagates local precision errors. The metric of interest is the relative terminal-state error

$$E_{\text{depth}} = \frac{\|x_L - \widehat{x}_L\|_{\text{HS}}}{\|x_L\|_{\text{HS}}}.$$

	Gaussian		Transformer	
Policy	depth 4	depth 8	depth 2	depth 4
`fp32_reference`	7.2e-8	8.8e-8	5.2e-8	2.2e-8
`bf16_safe`	3.3e-3	4.7e-3	3.4e-3	1.1e-3
`fp16_safe`	4.5e-4	6.7e-4	1.6e-4	1.7e-4
`bf16_low_logit`	3.3e-3	4.9e-3	3.4e-3	1.1e-3
`fp16_low_logit`	4.5e-4	6.3e-4	1.6e-4	1.7e-4
`bf16_low_softmax`	3.6e-3	4.7e-3	3.4e-3	1.1e-3
`fp16_low_softmax`	4.3e-4	6.8e-4	1.6e-4	1.7e-4
`bf16_low_value`	3.6e-3	5.3e-3	3.1e-3	1.2e-3
`fp16_low_value`	5.3e-4	7.8e-4	6.6e-4	6.1e-4

Table 14 / Relative terminal-state error E_{depth} for toy attention-only residual dynamical systems (no learned projections, MLP, or layer norm). Gaussian data, $n = d = 16$, averaged over 2 seeds. Depth $L = 4$ uses $h = 0.25$; depth $L = 8$ uses $h = 0.125$. The transformer column uses the same tiny-model smoke-test head as Table 12.

Table 14 reports E_{depth} for both Gaussian and transformer-activation residual stacks. On Gaussian data, error grows modestly with depth for all policies: the fp16-safe error roughly doubles from $L = 4$ to $L = 8$, while the bf16-safe error grows by roughly 50%. The low-precision value policies show the steepest depth growth, consistent with the $\sqrt{n}\|\Delta V\|_{\text{HS}}$ term in Proposition 3 accumulating across layers. On transformer activations, the residual scaling $h = 1/L$ keeps errors approximately constant or even decreasing with depth, reflecting the stabilizing effect of small residual contributions.

4.4 Ising Magnetization as a Learnability-Transition Test

To study whether reduced precision creates a training-time barrier, we add a toy Ising magnetization task. Each data point is an $L \times L$ spin configuration

$$X \in \{-1, +1\}^{L\times L},$$

sampled from the two-dimensional Ising model at a range of temperatures around the critical value $T_c \approx 2.269$. The normalized magnetization is

$$m(X) = \frac{1}{L^2} \sum_{i,j} X_{ij} \in [-1, 1],$$

and the regression target is the order parameter

$$y = |m(X)| \in [0, 1].$$

Each lattice site is treated as a token containing the spin value and its normalized two-dimensional position. A small one-head attention regressor ($d_{\text{model}} = 32$, depth 1) is trained to predict $|m(X)|$ via MSE loss.

This task is intentionally simple: a full model could compute magnetization by averaging. Its purpose is diagnostic rather than architectural. It gives a controlled setting in which microscopic spins must be aggregated into a macroscopic order parameter, while the data distribution itself changes sharply near the physical critical temperature.

Sanity check: full observation

When all spins are visible, every precision and sketch policy succeeds. Table 15 reports the result: all runs achieve $R^2 \geq 0.97$, and the learned attention remains uniformly diffuse ($H_{\text{attn}}/\log N \approx 1.0$, $M_{\text{attn}} \approx 1/N$). The unmasked task is a negative control: it validates the data pipeline and model implementation, but it is too easy to expose a precision-induced learnability barrier because uniform averaging is already sufficient.

Train policy	Eval policy	Test MSE	R^2	Success
`fp32`	`fp32`	2.8e-3	0.976	1
`fp16_safe`	`fp16_safe`	2.8e-3	0.975	1
`int8_qkv_dyn`	`int8_qkv_dyn`	2.9e-3	0.975	1
`sketch_4`	`sketch_4`	2.7e-3	0.977	1
`sketch_16`	`sketch_16`	3.0e-3	0.974	1
`fp32`	`fp16_safe_eval`	3.8e-3	0.967	1
`fp32`	`sketch_4_eval`	3.8e-3	0.967	1

Table 15 / Unmasked Ising magnetization. All policies succeed ($R^2 \geq 0.97$). The learned attention is uniformly diffuse, confirming that the task requires only global averaging.

Masked-spin magnetization prediction

To expose a learnability barrier, we randomly mask each spin with probability $1 - p$. The model receives

$$\widetilde{X}_{ij} = M_{ij}\, X_{ij}, \qquad M_{ij} \sim \text{Bernoulli}(p),$$

and each token encodes $[\tilde{s}_{ij},\ M_{ij},\ \text{row}_{ij},\ \text{col}_{ij}]$, so the model can distinguish "masked" from "physical spin zero." The target remains $y = |m(X)|$, computed from the *full* configuration.

We evaluate separately near the critical region $T \in [2.1, 2.5]$, where configurations fluctuate most and the inference problem is hardest. Success requires $R^2_{\text{near } T_c} \geq 0.80$ or $\text{MAE}_{\text{near } T_c} \leq 0.08$.

Table 16 reveals a finite-size learnability crossover. At $p = 0.8$ and $p = 0.4$, all policies succeed; the task is still easy enough that even aggressive sketching preserves the global aggregation. At $p = 0.2$, a barrier appears: the fp32 model trained from scratch achieves $R^2_{T_c} = 0.78$, just below the success threshold, while `sketch_4` trained from scratch reaches $R^2_{T_c} = 0.81$ and succeeds. Paradoxically, evaluating the trained fp32 model *under* sketch compression (`sketch_4_eval`) drops $R^2_{T_c}$ to 0.73, worse than training with

p	Train	Eval	R^2_{all}	$R^2_{T_c}$	MAE_{T_c}	Succ
0.8	`fp32`	`fp32`	0.967	0.935	0.047	1
0.8	`fp16_safe`	`fp16_safe`	0.968	0.936	0.046	1
0.8	`sketch_4`	`sketch_4`	0.968	0.924	0.051	1
0.4	`fp32`	`fp32`	0.941	0.895	0.060	1
0.4	`fp16_safe`	`fp16_safe`	0.940	0.895	0.060	1
0.4	`sketch_4`	`sketch_4`	0.934	0.882	0.061	1
0.2	`fp32`	`fp32`	0.874	0.778	0.085	0
0.2	`sketch_4`	`sketch_4`	0.892	0.814	0.080	1
0.2	`fp32`	`sketch_4_eval`	0.868	0.733	0.100	0
0.1	`fp32`	`fp32`	0.769	0.576	0.111	0
0.1	`sketch_4`	`sketch_4`	0.790	0.580	0.116	0
0.1	`fp32`	`sketch_4_eval`	0.754	0.501	0.135	0

Table 16 / Masked Ising magnetization prediction ($L = 12$, averaged over 2 seeds). $R^2_{T_c}$ and MAE_{T_c} are evaluated near the critical region. The full sweep is committed in `results/ising_masked_transition.dev.csv`.

the sketch from the beginning. At $p = 0.1$, all policies fail near T_c; the information deficit is too large for any attention policy to overcome.

The $p = 0.2$ regime is the most suggestive case in the current sweep. The fp32 model trained from scratch falls just below the chosen near-critical success threshold, while `sketch_4` trained from scratch crosses it. Conversely, applying the same sketch only at evaluation time hurts the fp32-trained model. This pattern is consistent with a training-time regularization effect, but it should not yet be interpreted as conclusive: the sweep uses only a small number of seeds, and the relevant metrics are close to the success boundary. A denser sweep over mask probabilities and more random seeds is needed to determine whether this becomes a robust learnability transition. The present Ising numbers should be read as a pilot experiment; the next version should estimate success probabilities over more seeds rather than relying on averaged metrics alone.

4.5 Approximation as a Training Barrier: A Kalman-Style Diagnostic

The masked Ising experiment suggests that reduced precision and randomized attention approximations should not be studied only as inference-time perturbations. During training, they also perturb the learning signal itself. We therefore add a diagnostic inspired by the predictor–corrector structure of the Extended Kalman Filter. The analogy is not literal: we do not maintain a covariance matrix over neural-network parameters. Instead, we use the full-precision attention step as a high-fidelity reference and the low-precision or sketched attention step as a noisy prediction.

For a fixed batch and fixed parameters, let

$$g = \nabla_\theta \mathcal{L}_{\text{fp32}}, \qquad \widehat{g} = \nabla_\theta \mathcal{L}_{\text{approx}}.$$

We measure the gradient cosine

$$\frac{\langle g, \widehat{g} \rangle}{\|g\|_2 \|\widehat{g}\|_2},$$

the relative gradient error

$$\frac{\|\widehat{g} - g\|_2}{\|g\|_2},$$

and the update signal-to-noise ratio

$$\mathrm{SNR}_{\mathrm{update}} = \frac{\|g\|_2}{\|\widehat{g} - g\|_2}.$$

When this SNR is below one, the numerical or randomized perturbation to the update is larger than the full-precision learning signal. This gives a concrete notion of a training barrier. We also compute the output innovation $\|\widehat{y}_{\mathrm{approx}} - \widehat{y}_{\mathrm{fp32}}\|_2$ and the relative output error.

We also test a simple Kalman-style correction schedule: most updates are performed under the approximate policy, but every few batches an fp32 update is inserted. This is a predictor–corrector analogue rather than a full EKF. Its purpose is to test whether occasional high-fidelity corrections help the model cross the approximation-induced training barrier.

Policy	p	$R^2_{T_c}$	MAE_{T_c}	Succ	Grad cos	Update SNR	Grad rel err
`bf16_safe`	0.2	0.649	0.106	0.0	1.000	1526	0.001
`fp16_safe`	0.2	0.632	0.113	0.0	1.000	2128	0.000
`int8_logits_dyn`	0.2	0.671	0.112	0.0	1.000	56	0.020
`int8_qkv_dyn`	0.2	0.607	0.115	0.0	0.583	1.23	0.811
`sketch_4`	0.2	0.686	0.105	0.0	0.083	0.92	1.335
`sketch_8`	0.2	0.579	0.120	0.0	0.126	0.96	1.391
`bf16_safe`	0.3	0.721	0.099	0.5	1.000	1367	0.001
`fp16_safe`	0.3	0.761	0.090	0.0	1.000	1167	0.001
`int8_logits_dyn`	0.3	0.767	0.091	0.5	1.000	62	0.023
`int8_qkv_dyn`	0.3	0.685	0.105	0.0	0.607	1.26	0.794
`sketch_4`	0.3	0.757	0.092	0.0	0.485	1.22	0.843
`sketch_8`	0.3	0.742	0.096	0.0	0.240	1.42	1.221
`bf16_safe`	0.4	0.803	0.083	0.5	1.000	708	0.002
`fp16_safe`	0.4	0.808	0.080	0.5	1.000	934	0.001
`int8_logits_dyn`	0.4	0.779	0.092	0.5	1.000	49	0.022
`int8_qkv_dyn`	0.4	0.761	0.089	0.5	0.617	1.27	0.786
`sketch_4`	0.4	0.822	0.079	1.0	0.044	0.87	1.257
`sketch_8`	0.4	0.799	0.084	0.5	0.838	1.36	0.872

Table 17 / Training-barrier diagnostics on masked Ising magnetization ($L = 12$, 15 epochs, averaged over 2 seeds). Gradient metrics compare each approximate policy against an fp32 reference on the same probe batch. The full sweep is committed in `results/ising_barrier_diagnostics.dev.csv`.

Table 17 reports the gradient diagnostics at the final training epoch. The policies separate into two groups. The floating-point policies (`bf16_safe`, `fp16_safe`, `int8_logits_dyn`) have gradient cosine ≈ 1.0, update SNR $\gg 1$, and relative gradient error ≤ 0.02: their approximate gradients are nearly aligned with the fp32 reference. The sketch policies (`sketch_4`, `sketch_8`) have gradient cosine $\approx$ 0–0.5, update SNR $\approx$ 0.9–1.4, and relative gradient error > 0.8: the sketch noise is comparable to or larger than the learning signal. The `int8_qkv_dynamic` policy falls between these groups, with cosine ≈ 0.6 and SNR just above one. Despite the gradient misalignment, `sketch_4` at $p = 0.4$ achieves $R^2_{T_c} = 0.82$ and succeeds, while `fp16_safe` at $p = 0.3$ with perfectly aligned gradients achieves only $R^2_{T_c} = 0.76$. This confirms that gradient alignment is informative about the nature of the approximation error, but does not by itself predict training success: a misaligned gradient can still move the parameters in a useful direction.

Train mode	p	$R^2_{T_c}$	MAE_{T_c}	Succ	Final grad cos
`fp32`	0.2	0.634	0.113	0.0	1.000
`sketch_4`	0.2	0.650	0.108	0.0	−0.865
`sketch_4_periodic_fp32_correction`	0.2	0.595	0.117	0.0	0.091
`sketch_8`	0.2	0.653	0.109	0.0	0.029
`sketch_8_periodic_fp32_correction`	0.2	0.626	0.113	0.0	0.931
`int8_logits_dyn`	0.2	0.676	0.104	0.0	1.000
`int8_logits_periodic_fp32_correction`	0.2	0.691	0.105	0.0	1.000
`fp32`	0.3	0.762	0.090	0.0	1.000
`sketch_4`	0.3	0.806	0.084	0.5	0.042
`sketch_4_periodic_fp32_correction`	0.3	0.725	0.099	0.0	0.655
`sketch_8`	0.3	0.794	0.085	0.5	0.373
`sketch_8_periodic_fp32_correction`	0.3	0.771	0.090	0.5	0.985
`int8_logits_dyn`	0.3	0.738	0.095	0.5	1.000
`int8_logits_periodic_fp32_correction`	0.3	0.766	0.089	0.5	1.000
`fp32`	0.4	0.809	0.080	0.5	1.000
`sketch_4`	0.4	0.856	0.070	1.0	0.974
`sketch_4_periodic_fp32_correction`	0.4	0.789	0.086	0.5	0.900
`sketch_8`	0.4	0.840	0.077	1.0	0.873
`sketch_8_periodic_fp32_correction`	0.4	0.768	0.090	0.5	0.971
`int8_logits_dyn`	0.4	0.815	0.080	0.5	1.000
`int8_logits_periodic_fp32_correction`	0.4	0.793	0.085	0.5	0.999

Table 18 / Kalman-style full-precision correction schedule ($L = 12$, 15 epochs, averaged over 2 seeds). Approximate policies are trained from scratch, with optional fp32 correction steps every five batches. The full sweep is committed in `results/ising_correction_sweep.dev.csv`.

Table 18 reports the correction-sweep results. The correction schedule does not substantially improve the sketch policies in this pilot sweep. At $p = 0.2$, where all policies fail, adding periodic fp32 corrections to `sketch_4` actually hurts ($R^2_{T_c}$ drops from 0.650 to 0.595), though it improves the final gradient cosine from -0.865 to 0.091. For `sketch_8`, correction improves the gradient cosine (0.029→0.931) without improving $R^2_{T_c}$. At $p = 0.4$, where `sketch_4` already succeeds, correction again hurts ($R^2_{T_c}$ drops from 0.856 to 0.789). For the int8 policies, which already have well-aligned gradients, correction produces negligible changes. This suggests that the observed failures are not caused simply by occasional misaligned updates, or that the correction frequency of one-in-five batches is insufficient. The gradient diagnostics nevertheless remain useful because they quantify whether the approximate policy follows the same local learning direction as fp32, and they cleanly separate the floating-point policies (cosine≈ 1) from the sketch policies (cosine≈ 0).

4.6 A Stress-Test Baseline

The experiments in this subsection probe the decomposition of Theorem 9 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

(1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);

(2) $\widetilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;

(3) $\widehat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\text{sk}} := \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}}$ and the finite-precision error is $E_{\text{fp}} := \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}$. The total error is $E_{\text{tot}} := \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\text{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (storage, projection, final logit product, and softmax in float16), `mixed_fp16` (storage and projection in float16, followed by a float32 final logit product and float32 softmax), and `fp8_store` (storage and projection in float8 E4M3, followed by a float32 final logit product and float32 softmax).

The sketch $S \in \mathbb{R}^{d\times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Since the synthetic Q and K are i.i.d. Gaussian, their joint row space is typically full-dimensional, so for $d = 64$ and $s \leq 32$ the subspace-embedding regime from Proposition 6 is not expected to hold. The experiment should therefore be read as an intentionally under-sketched stress test of the error decomposition, not as evidence that such small feature sketches preserve attention accurately.

Table 19 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Three observations stand out. First, for fp32 and fp16, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3–10^6), confirming that the dominant error source is the randomized sketch. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent with Proposition 7: storage quantization dominates any benefit from higher-precision accumulation. Third, fp8 storage substantially increases the finite-precision error (from $\sim$0.3 to $\sim$30 in absolute terms), but the sketching error still dominates by a factor of $\sim$14–15. The results support the error decomposition and illustrate the degradation predicted by Theorem 9 for increasing $s \cdot u$, but they do not demonstrate a high-quality efficient-attention approximation or a full inversion of the error ratio.

5 Outlook

The main contribution of this report is the precision-placement decomposition of Proposition 3 and its empirical validation: low-precision softmax is the clearest failure mode in exact attention, while low-precision storage of Q, K, V with fp32 accumulation is typically safe. The sketching baseline (Theorem 9) confirms that the finite-precision term remains lower order as long as $s \cdot u \ll \epsilon$. Mixed precision and quantization are both precision-placement mechanisms, but quantization adds calibration and clipping

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
`fp32`	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
`full_fp16`	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
`mixed_fp16`	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
`fp8_store`	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 19 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds). $E_{\text{sk}} = \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

error; the same local perturbation framework applies to both, provided the quantized tensor is modeled as $X + \Delta X$, where ΔX is controlled not only by unit roundoff but also by scale, zero-point, range selection, and saturation.

Several directions remain open. First, the current experiments use small problem sizes ($n \leq 256$, $d \leq 64$); scaling to realistic transformer dimensions will require either GPU acceleration or the JAX backend scaffold already present in the repository. Second, FP8 policies are reported only under an explicit quantization model; a more complete treatment would include hardware-specific scaling and dequantization. Third, the phase-diagram analogy from Decelle et al. [8] suggests mapping not only average error but also regime changes: when does low-precision logit storage remain harmless, when does low-precision softmax begin to amplify, and when does no practical low-precision policy preserve the reference behavior? The current data already hints at these transitions but does not yet chart them systematically. Fourth, the JAX-focused mixed-precision framework of Graefe and Trimpe [9] provides a natural implementation path for the safe/unsafe policy distinctions identified here. We note that the current JAX backend in the repository maps PyTorch fp8 dtypes to JAX float16, so JAX-based fp8 experiments would require a dedicated quantization path and should not be compared directly with PyTorch fp8 results. Fifth, the masked Ising experiment suggests a finite-size learnability crossover near $p \approx 0.2$: below this mask probability the task becomes difficult near T_c, while above it all tested policies succeed. The apparent advantage of training with the sketch from the beginning over applying it only after fp32 training is suggestive of a training-time regularization effect, but it should be checked with more seeds, denser mask probabilities, and larger model variants.

The Ising barrier diagnostics suggest a broader direction: mixed precision and RandNLA should be analyzed not only through forward error, but also through the distortion they introduce into the training signal. A more complete theory would bound the difference between full-precision and approximate gradients, and relate this difference to whether training can cross the entropy-collapse or aggregation transition of the attention layer. The predictor–corrector analogy with Kalman filtering is a useful practical diagnostic: approximate updates can be viewed as noisy predictions, while occasional full-precision updates act as high-fidelity corrections. In the current pilot sweep, the correction schedule does not substantially improve the sketch policies, but the gradient diagnostics cleanly separate floating-point policies (gradient cosine≈ 1) from sketch policies (gradient cosine≈ 0), confirming that the two classes of approximation distort the learning signal in qualitatively different ways.

On the analytical side, the $\sqrt{n}$ bound on $\|\Sigma(\widehat{L})\|_2$ is likely loose for near-doubly-stochastic attention matrices; a refined bound that exploits the concentration of softmax rows could tighten the value-perturbation term. The logit-perturbation term could similarly be refined by replacing the worst-case $\|V\|_2$ with an average-case or data-dependent quantity.

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